Robotics



Mahmoud Abdelaal

Lecture # 3

Sensors for Robotic Systems I

Sensors for Robotic Systems

Definition

- The term **sensor** is used for an input device that provides a usable output in response to a specified physical input.
- **For example**, a thermocouple is a sensor which converts a temperature difference into an electrical output.

The General Question

- Given the sensory reading what was the world like?

Example: Skin

Sensors

- Simple ←→ complex

Sensors for Robotic Systems

Types of Sensors

- Contact: switch
- Distance: Ultrasound, radar, infra red
- Light level: photo cells, cameras
- Sound level: microphone
- Strain: strain gauge
- Rotation: encoders
- Smell: chemical
- Temperature: thermal, infra red
- Inclination: inclinometers, gyroscopes
- Pressure: pressure gauges
- Altitude: altimeters

Sensors for Robotic Systems

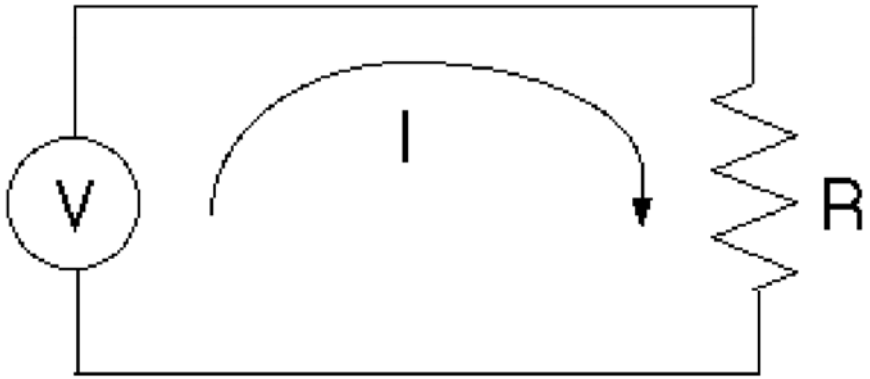
Levels of Processing

- A switch:
  - open = 0 volts
  - Closed = 5 volts
- A digital scale:
- Microphone:
- Camera:

Sensors for Robotic Systems

Ohm's Law

- V= IR
  - V =voltage (volts)
  - I =current (Amps)
  - R = resistance (Ohms)

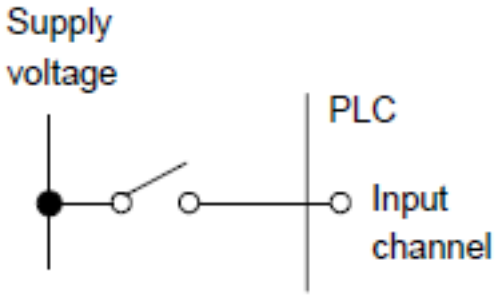


Mechanical switches / Limit Switches

Mechanical switches / Limit Switches

Mechanical switches

- A mechanical switch generates an on-off signal or signals as a result of some mechanical input causing the switch to open or close.
- Ex. the presence of a workpiece on a machining table
- The absence of the workpiece is indicated by the switch being open and its presence by it being closed.
- The 1 level might correspond to a 24 V DC input, the 0 to a 0 V input.



Mechanical switches / Limit Switches

Mechanical switches

- The term limit switch is used for a switch which is used to detect the presence or passage of a moving part. It can be actuated by a cam, roller or lever.

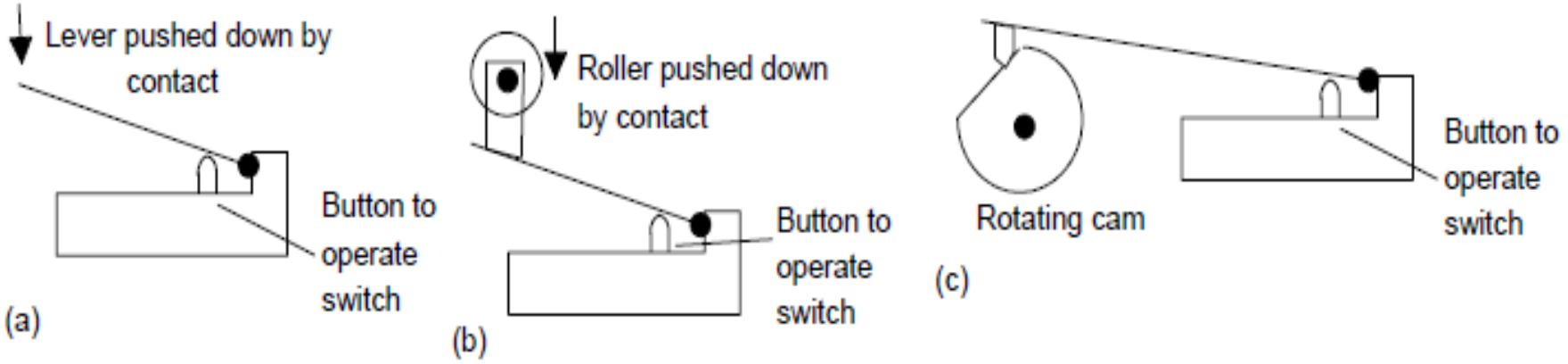


Figure: Limit switches actuated by: (a) lever, (b) roller, (c) cam

Mechanical switches / Limit Switches

Limit Switches

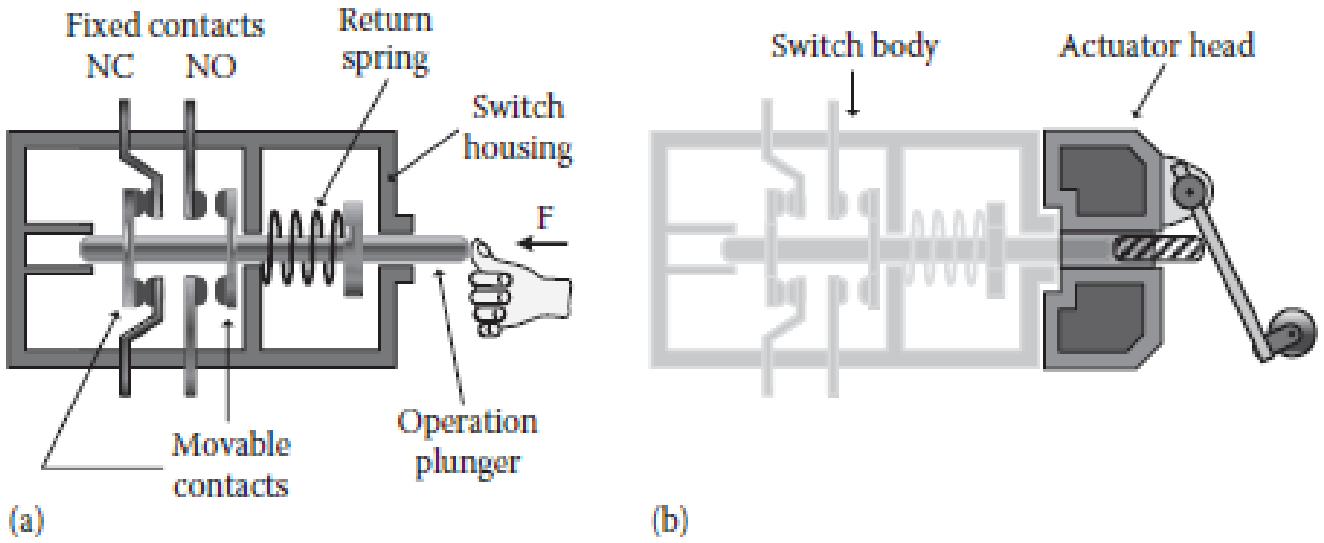
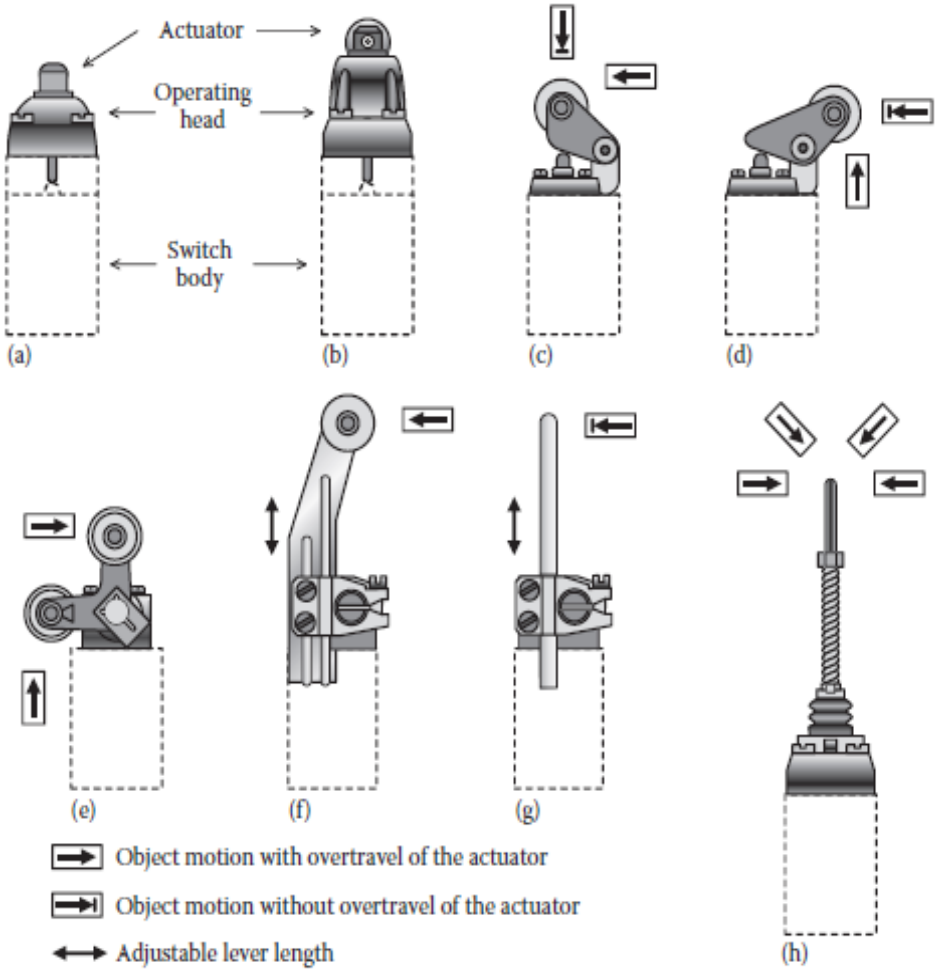


Figure: Typical form of a limit switch with (a) an interior view of switch body, and (b) a switch body equipped with an actuator head.

Mechanical switches / Limit Switches

Limit Switches

- The only difference being the type of sensor head.
- Figure: Various types of actuator head for transmitting the external force to the changeover mechanism suitably and thereby engaging the movable contact's
- (a) push plunger,
- (b) roller plunger,
- (c) roller lever plunger (motion from the front or side),
- (d) roller lever plunger (motion from the back or side),
- (e) fork type,
- (f) adjustable roller arm,
- (g) adjustable rod, and
- (h) spring flexible rod.



Mechanical switches / Limit Switches

Limit Switches

Advantages

- Limit switches are cheaper in comparison to proximity or photoelectric switches
- There is also a wide range of limit switches regarding their size.

Proximity Switches

Proximity Switches

Proximity Switches

- There are three basic types of proximity switches with respect to their operating principles: inductive, capacitive, and magnetic.
- The operation of inductive proximity switches is based on the variation of the magnetic inductance.
- The capacitive proximity switches base their operation on the variation of capacitance and the magnetic proximity switches on the variation of the magnetic flow.
- The most widely-used proximity switches in industrial applications are the inductive and the capacitive ones.



Figure: Typical external view of a proximity sensor.

Proximity Switches

Proximity Switches

- Proximity switches are used to detect the presence of an item without making contact with it.
- There are a number of forms of such switches, some being only suitable for metallic objects.
- The voltage amplitude is thus a measure of the proximity of metallic objects.
- The voltage can be used to activate an electronic switch circuit, basically a transistor which has its output switched from low to high by the voltage change, and so give an on-off device.
- The range over which such objects can be detected is typically about 0.5 to 20 mm.

Proximity Switches

Capacitive Proximity Switch

Proximity Switches

Capacitive Proximity Switch

- A proximity switch that can be used with metallic and non-metallic objects is the capacitive proximity switch.
- The capacitance of a pair of plates separated by some distance depends on the separation, the smaller the separation the higher the capacitance.
- The sensor of the capacitive proximity switch is just one of the plates of the capacitor, the other plate being the metal object whose proximity is to be detected.

Proximity Switches

Capacitive Proximity Switch

- The sensor can also be used to detect non-metallic objects since the capacitance of a capacitor depends on the dielectric between its plates.
- In this case the plates are the sensor and the earth and the non-metallic object is the dielectric.
- The change in capacitance can be used to activate an electronic switch circuit and so give an on-off device.
- Capacitive proximity switches can be used to detect objects when they are typically between 4 and 60 mm from the sensor head.

Proximity Switches

Proximity Switches

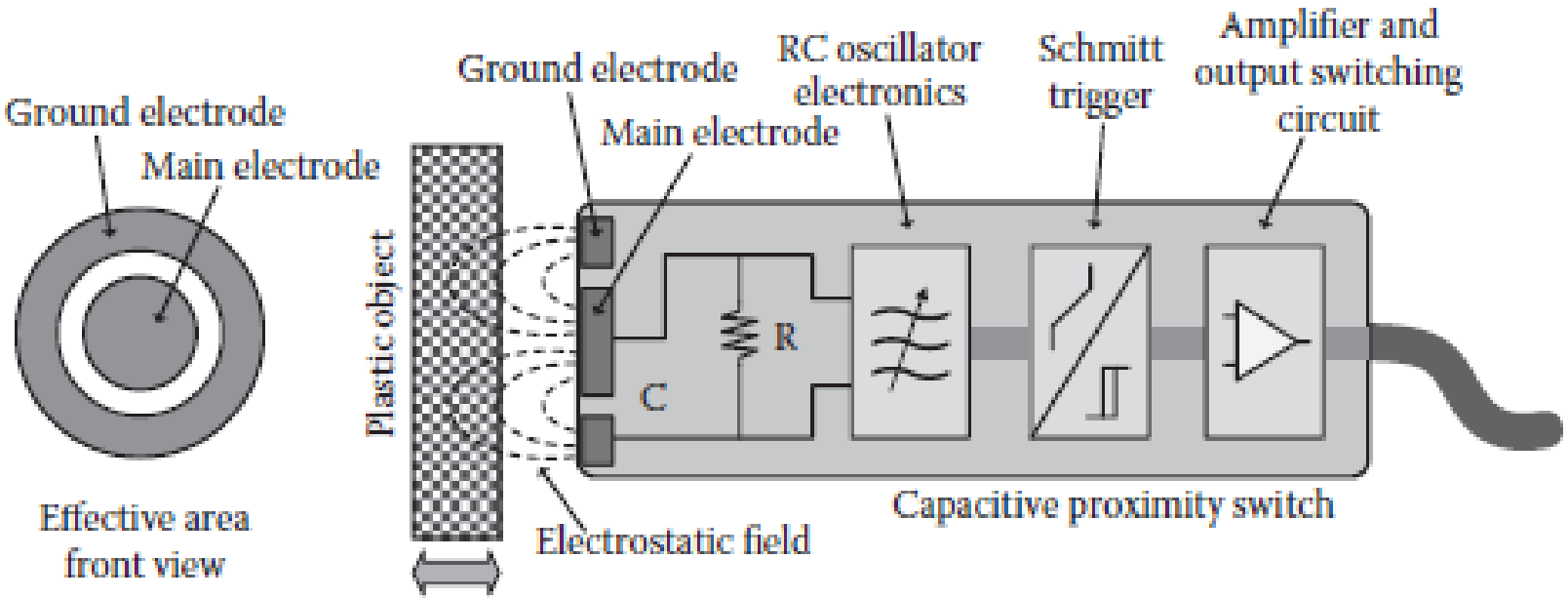


Figure: Internal elements of a capacitive proximity sensor and object-affecting dielectric constant.

Proximity Switches

Capacitive Proximity Switch

Proximity Switches

Inductive Proximity Switch

- Another type, the inductive proximity switch, consists of a coil wound round a ferrous metallic core.
- When one end of this core is placed near to a ferrous metal object there is effectively a change in the amount of metallic core associated with the coil and so a change in its inductance.
- This change in inductance can be monitored using a resonant circuit, the presence of the ferrous metal object thus changing the current in that circuit.
- The current can be used to activate an electronic switch circuit and so give an on-off device.
- The range over which such objects can be detected is typically about 2 to 15 mm.

Proximity Switches

Proximity Switches

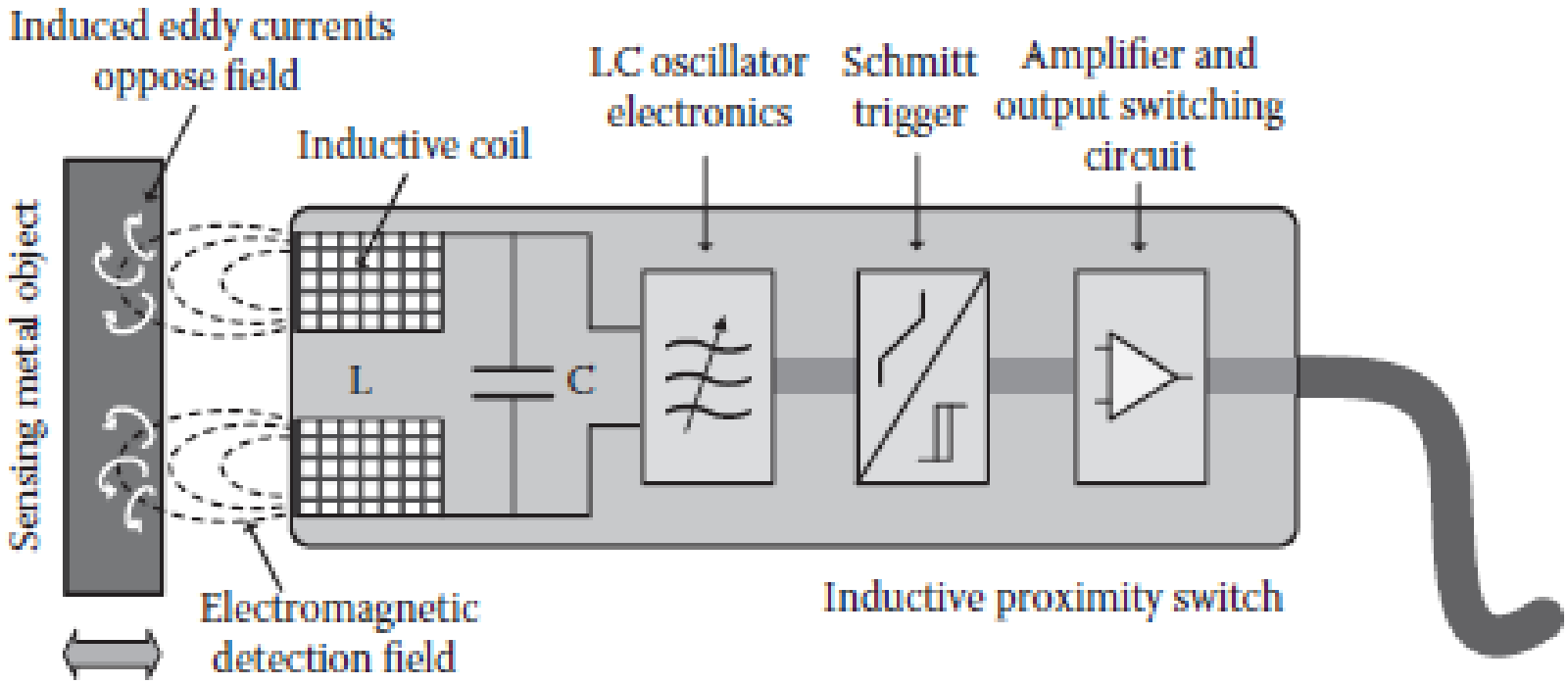
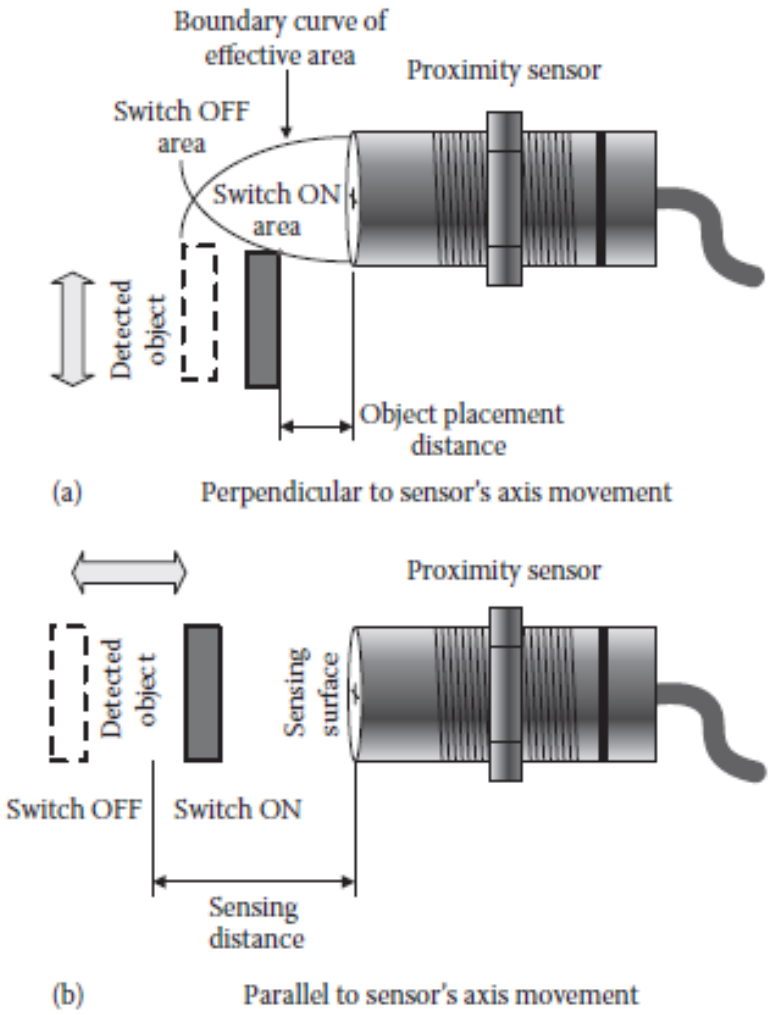


Figure: Internal elements of an inductive proximity sensor and object-electromagnetic field interaction.

Proximity Switches

Inductive Proximity Switch

- Figure: Object movement in front of the sensing area of a proximity sensor perpendicular to the sensor's longitudinal axis (a), and parallel to the sensor's longitudinal axis (b).



Proximity Switches

Proximity Switches

- The main types of proximity switches, with respect to their external form and shape, are the cylindrical, rectangular, and the slot ones.

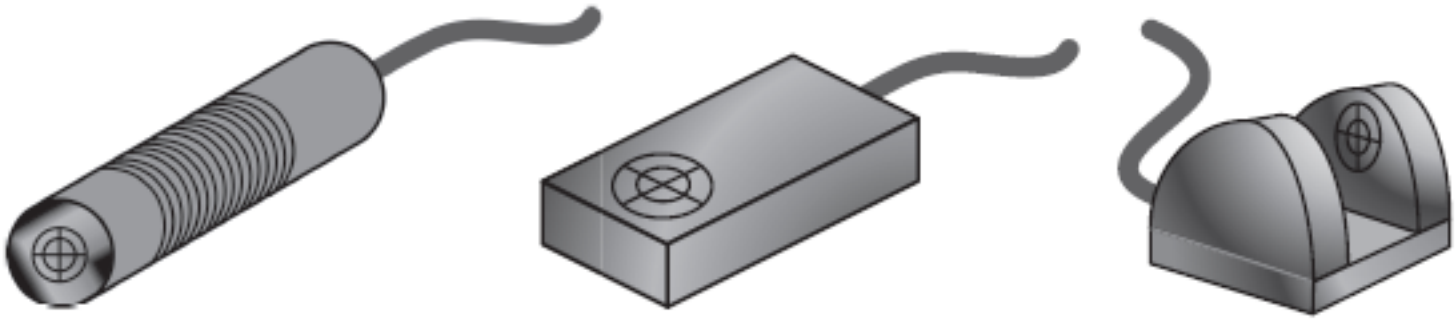


Figure: Basic exterior forms of proximity sensors for various applications.

Proximity Switches

Proximity Switches

- Electric connection of proximity sensors:
- Inductive and capacitive proximity switches can be connected to an AC or DC power supply source in a series or separately with a load, the so-called two-wire or three-wire connections.
- The connected load may be the coil of a power relay, the electronic circuit of a control unit, or the digital input of a programmable logic controller.

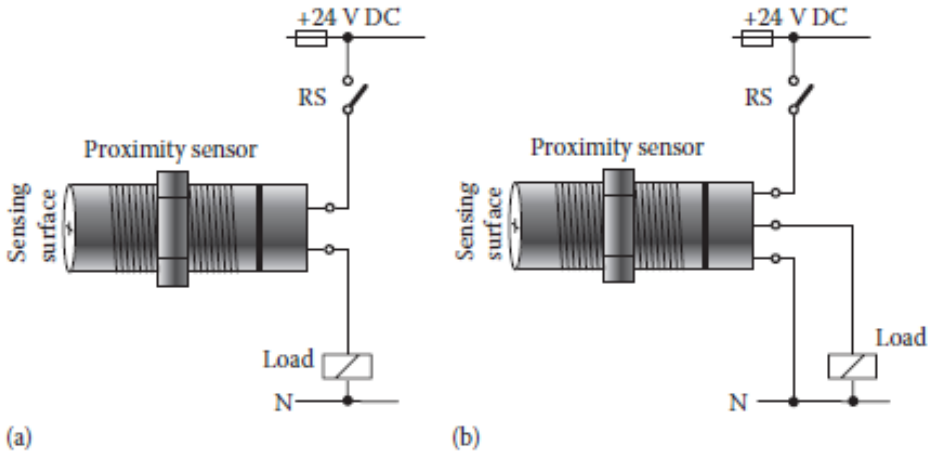


Figure: Electric circuit of a proximity switch with (a) a two-wire connection, and (b) a three-wire connection.

Proximity Switches

Proximity Switches

The advantages of using proximity switches over other detectors (e.g., limit switches) are as follows:

- Long electrical life.
- Durability (able to last a long time without being damaged) to vibrations, accelerated motion, and toxic environments.
- Operation under DC or AC voltage.
- Effective response in objects' high speed approaching and high frequency excitation.
- Absence of any kind of movable elements.

Proximity Switches

- **Proximity Switches**

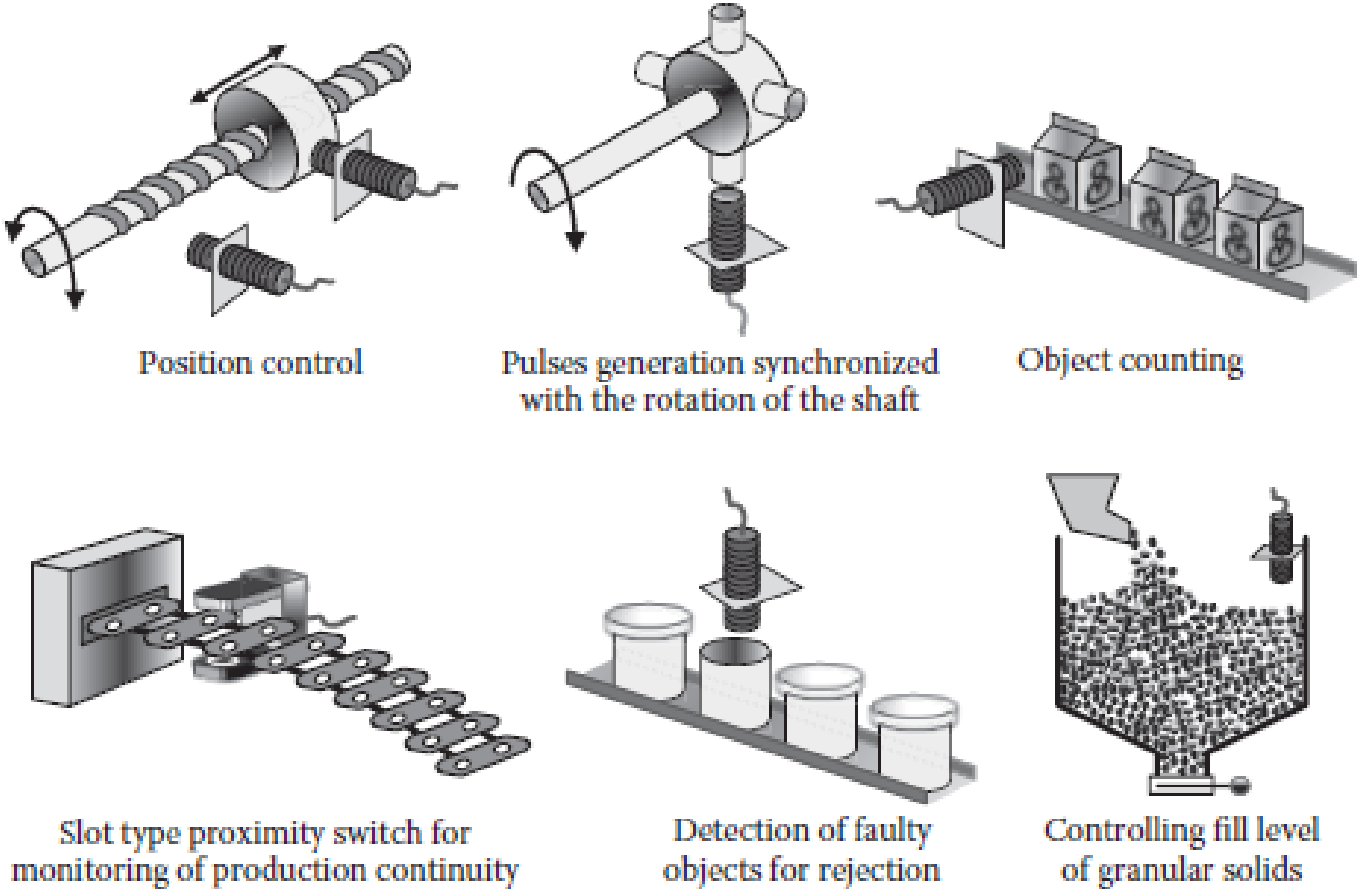


Figure: Typical applications of proximity switch usage.

Photoelectric sensors and switches

Photoelectric sensors and switches

Photoelectric sensors and switches

- Photoelectric switches, called also photocells, are solid-state sensors that are able to detect the presence of an object at long ranges.
- The non-contact detection is performed by a beam of light, visible or infrared, which can be interrupted or ignored by the detected object.
- In general, there are three basic types of photoelectric switches, regarding the utilized reflection type of the beam, which can be categorized as the **"through-beam", "retro-reflective", and "diffuse"**.



Figure: Typical exterior view of a photoelectric sensor.

Photoelectric sensors and switches

Through-beam photoelectric switches:

- Any photoelectric switch consists of an emitter and a receiver of the light beam.

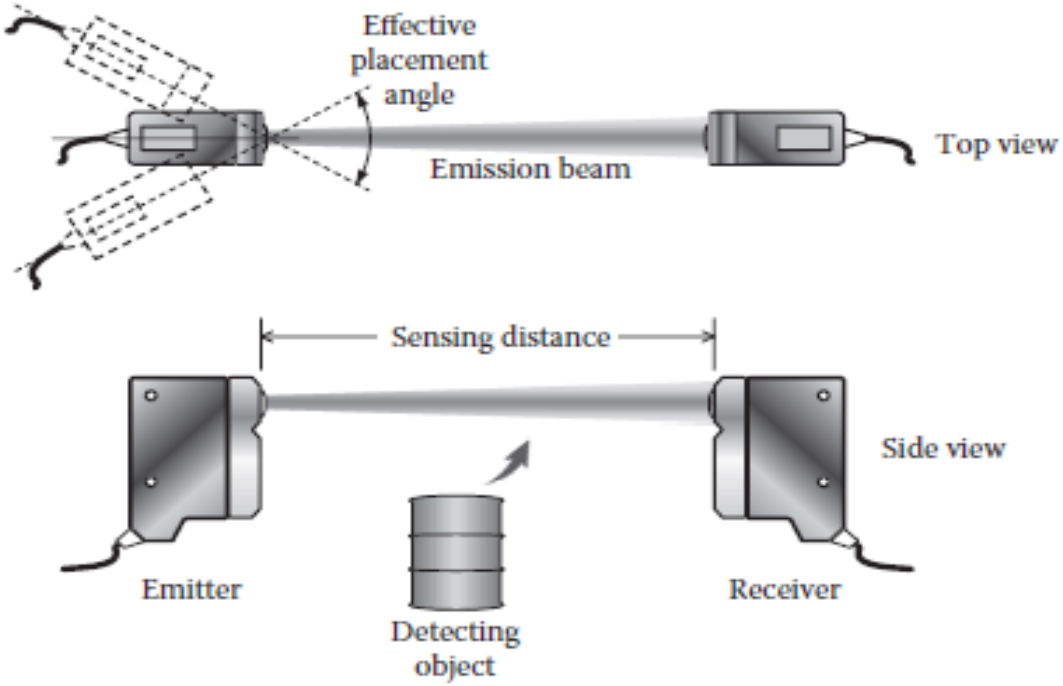


Figure: Through-beam photoelectric switch.

Photoelectric sensors and switches

Through-beam photoelectric switches:

- In a through-beam type of photoelectric switch, the emitter and the receiver are separate units, contained in different housings and positioned opposite each other.
- The sensing distance is the maximum operating distance between the emitter and receiver, and is achieved when the two units have perfect alignment and hence the maximum amount of the emitted energy reaches the receiver.
- This type of photoelectric switch has the advantage of a longer sensing distance, in comparison to other types of photoelectric switches.

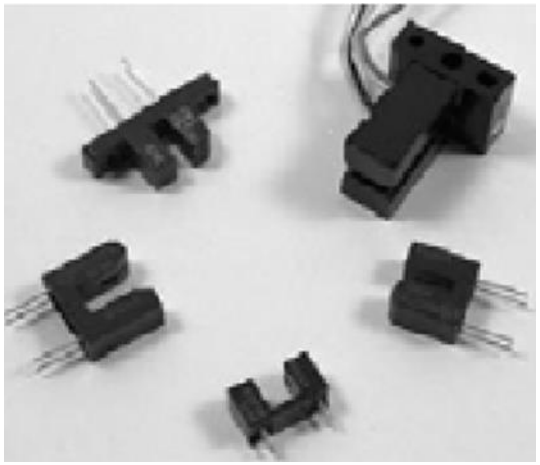
Photoelectric sensors and switches

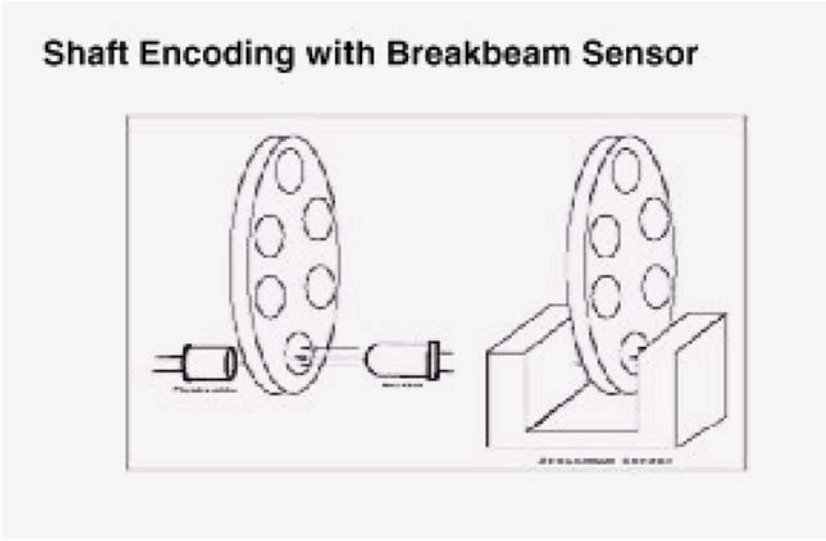
Or Break Beam Sensors

- Any pair of compatible emitter-detector devices can be used to make a break-beam sensor

Examples:

- Where have you seen these?





Photoelectric sensors and switches

Retro-reflective photoelectric switches:

- In this type of photoelectric switch, the **emitter** and **receiver** are **embedded in the same housing unit**, and a **reflector** is used for the reflection of the light beam.
- Reflectors are usually made up of many **small**, corner-cube prisms that reflect the beam back to the receiver, almost in parallel to the entering beam.
- A **simple mirror** can also be used as reflector.

Photoelectric sensors and switches

Retro-reflective photoelectric switches:

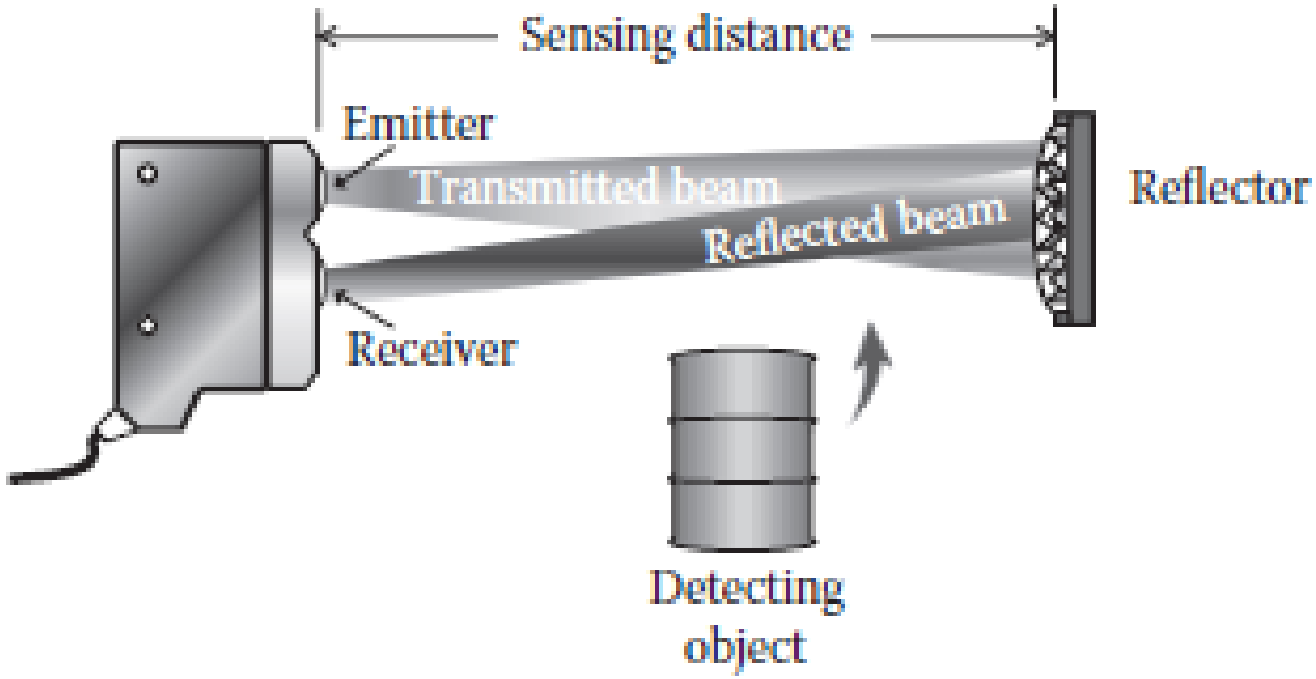


Figure: Retro-reflective photoelectric switch.

Photoelectric sensors and switches

Diffuse-reflective photoelectric switches,

- which base their operation on the **reflection** of the light beam **directly** on the surface of the **detected object**, are also called **"direct reflection"** photoelectric switches for this reason.
- Obviously, in this case the **emitter** and the **receiver** are embedded together in **one housing** unit, as in the retro-reflective type.
- The emitter emits a beam of light that **is not returned** by reflection to the receiver.
- When the target object is **inserted** in the light beam trajectory, the beam is **diffused** in many directions, one of which is **reflected back to the receiver**.

Photoelectric sensors and switches

Diffuse-reflective photoelectric switches

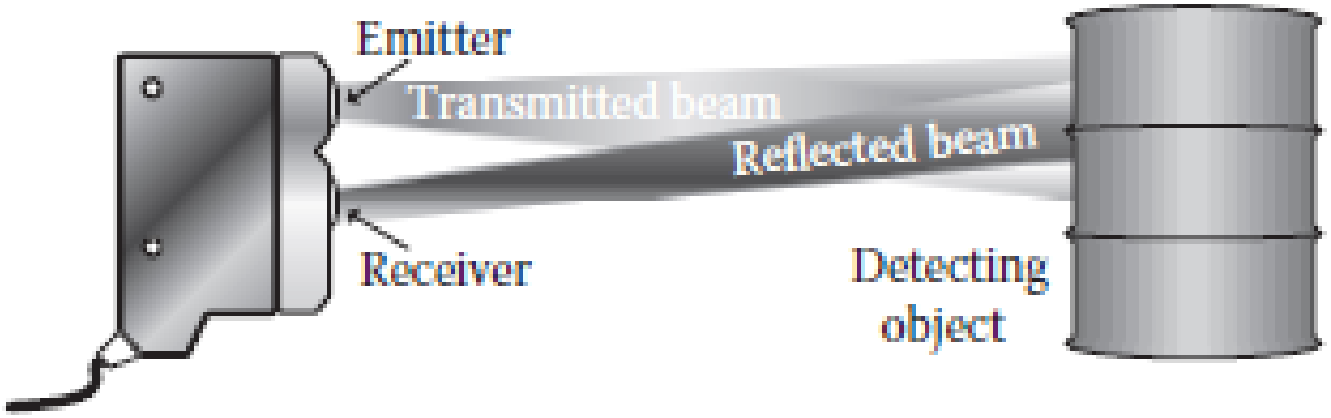


Figure: Diffuse-reflective photoelectric switch.

Photoelectric sensors and switches

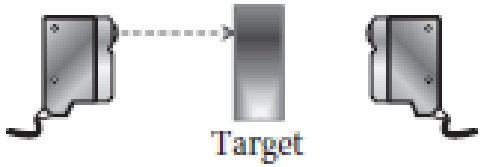
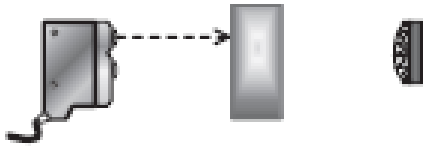
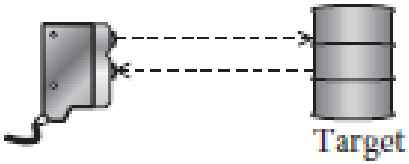
Photoelectric sensors and switches

- The detection **distance varies** with the **type**, **size**, and **model** of the photoelectric switch and ranges usually from less than **10 cm up to 1500 cm**.
- A new term, used only in photoelectric switches, refers to their **"Dark On"** or **"Light On"** operations.
- The **"Dark On"** operation means that the photoelectric switch is **energized** when the beam of light is **interrupted**, or simply when the switch **is "in the dark"**.
- Instead, the **"Light On"** operation means that the switch is **energized** when the beam of light **reaches** the receiver or simply when the switch is **"in the light"**.

Photoelectric sensors and switches

Photoelectric sensors and switches

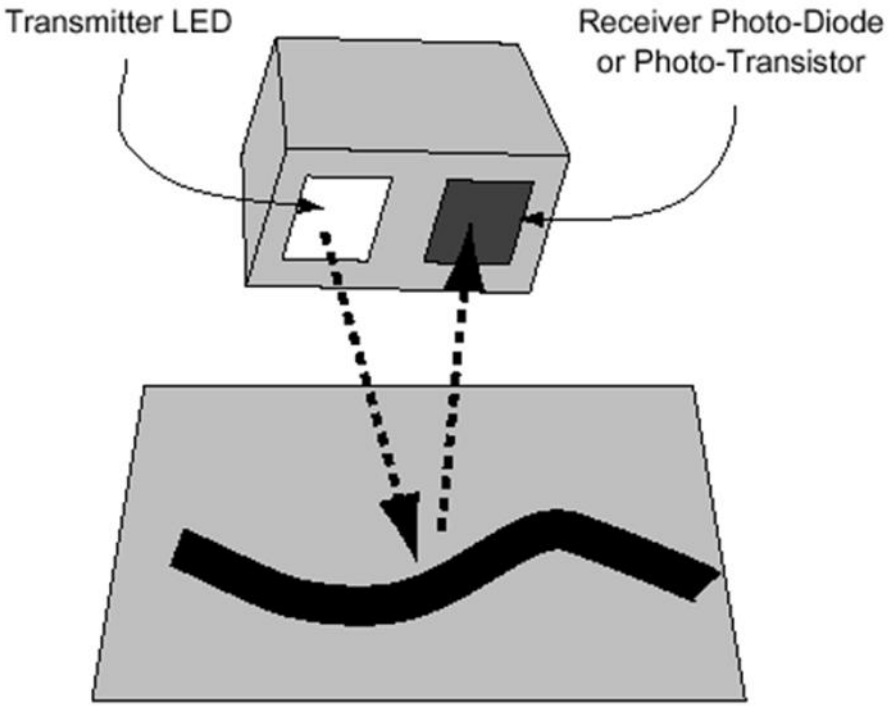
- The Dark-on and Light-on Operation of a Photoelectric Switch

Photoelectric Switch Type	*Light ON*	*Dark ON*
Through beam		
Retro-reflective		
Diffuse reflective		

Photoelectric sensors and switches

Reflective Opto-sensors

- Emitter and detector
- Emitter:
  - LED
- Detector:
  - Photodiode
  - Phototransistor



Photoelectric sensors and switches

Applications

- object presence detection
- object distance detection
- surface feature detection (finding/following markers/tape)
- wall/boundary tracking
- rotational shaft encoding

Photoelectric sensors and switches

Sensor limitations

- Light reflectivity:
  - Surface color
  - Texture
- Ambient light
- Sensor calibration

Light Sensors

Light Sensors

Light Sensors

A variable resistor that changes based on the light.

- Brighter light ➔ low resistance
- Darker light ➔ Higher resistance



- Note: Shielding, position, and directionality of the photocells are important.

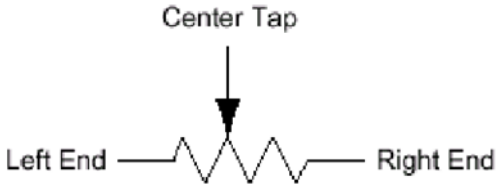


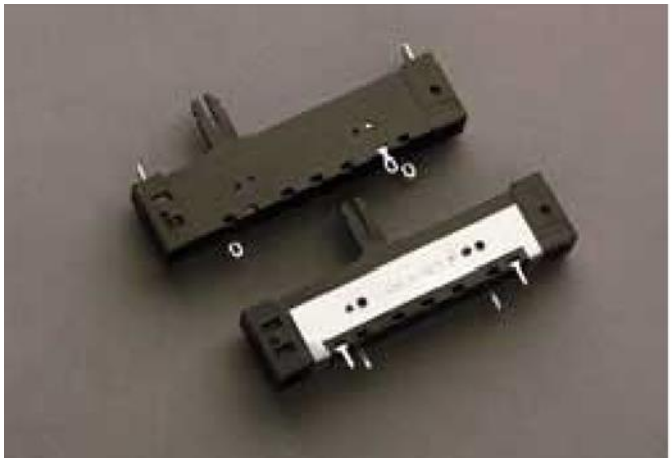
Potentiometers

Potentiometers

Potentiometers

- Volume control in your stereo
- Typically called pots

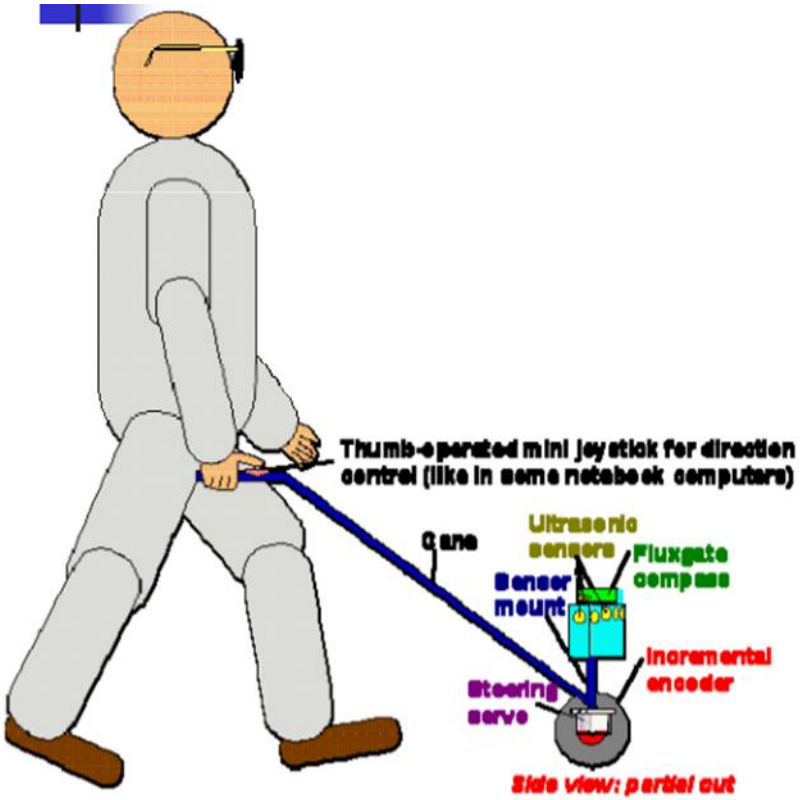






GuideCane Application

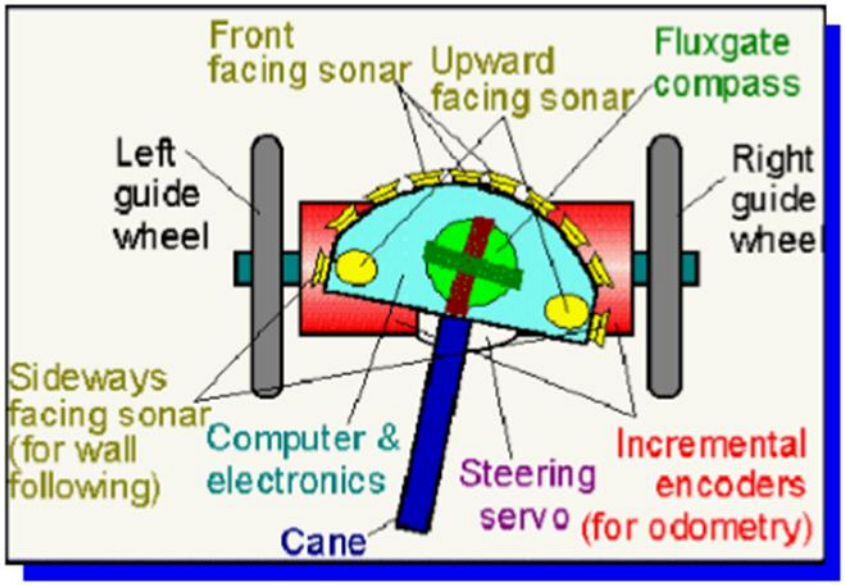
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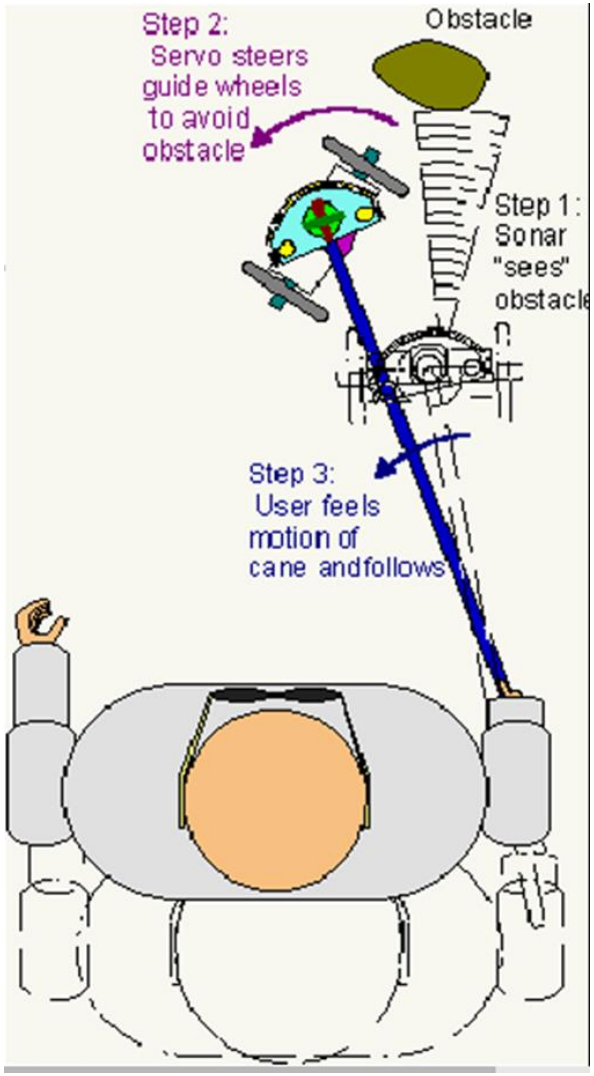


GuideCane Application

GuideCane



GuideCane Sensor Head (Top View)



Thank you for your attention !

Good Luck !

